# BHAGWAN MAHAVIR UNIVERSITY

## Bhagwan Arihant Institute of Technology, Surat.

A project on

# “SpendSense”

**Financial Transaction Anomaly Detection & Audit Platform**

Under the Subject of
**Project (2010206721)**

**B.Tech-IV, Semester-VII**
(Computer Engineering)


**Submitted by :**

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**Academic Year**
(2025-2026)

# Bhagwan Arihant Institute of Technology, Surat.

## CERTIFICATE

This is to certify that the project entitled **“SpendSense”** has been carried out by **Dhyan Surani (2407020603068)**, **Bhumil Sojitra (2407020603067)**, **Yash P. Patel (2407020603046)**, **Yug Radadiya (2407020603052)**, and **Siddh Patel (2407020603045)** at **Bhagwan Arihant Institute of Technology** under my guidance and supervision in fulfillment of the subject **PROJECT (2010206721) - 7th Semester**, Degree of Bachelor of Technology in Computer Engineering of Bhagwan Mahavir University, Surat during the academic year 2025-26.

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This is to certify that project Entitled

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In partial fulfilment of the Project (2010206721) in the Computer Engineering Department of Bhagwan Arihant Institute of Technology, Surat, is hereby approved submitted project for the 7th semester.

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# DECLARATION

We hereby declare that the project work entitled **“SPENDSENSE”** submitted by us to **BMU, SURAT**, in partial fulfilment of the requirement for the award of the degree of **B.Tech. Computer Engineering** is a record of bonafide project work carried out by us under the guidance of **Prof. Abhay Hadiya**. We further declare that work reported in this project will not be submitted, either in part or in full, for the award of any degree in this institute or any other institute or university.

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Apart from individual efforts, the success of this project depends largely on the encouragement and support of many others. We take this opportunity to express our sincere gratitude to everyone who has supported and guided us in making this project a reality.

We are heartily thankful to **Bhagwan Arihant Institute of Technology** for providing the conducive academic environment and computing resources. We deeply owe our gratitude to our project guide and coordinator **Prof. Abhay Hadiya** for continuous technical guidance, insightful suggestions, and supervision from the preliminary ideation stage to the final deployment level.

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# ABSTRACT

This project focuses on leveraging unsupervised machine learning algorithms, statistical boundaries, and high-throughput in-memory architecture to identify fraudulent, suspicious, and outlying financial transactions in large datasets. In modern accounting, audit, and commercial workflows, financial analysts face immense volumes of data where illicit patterns, duplicates, anomalous charge amounts, and data entry discrepancies are hidden amidst thousands of normal records. Traditional rule-based validation systems fail to adapt to dynamic behavioral shifts, generate high rates of false positives, and introduce heavy computational lag when processing large-scale transaction batches.

To address these operational bottlenecks, **SpendSense** provides an end-to-end, high-performance anomaly detection platform built on FastAPI (Python) and Next.js (React/TypeScript), integrated with **Supabase Cloud** for secure multi-tenant authentication and team workspace management. SpendSense combines an unsupervised **Isolation Forest** ensemble algorithm paired with **Interquartile Range (IQR)** statistical thresholds to score records without requiring labeled historical training sets. The platform features automated CSV schema mapping, chunked streaming ingestion with real-time multi-stage memory progression trackers, interactive visual analytics via Recharts, and a human-in-the-loop audit workflow allowing auditors to exclude, verify, or re-include flagged records with transparent, identity-signed audit trails.

To maximize enterprise utility while remaining ideal for evaluation, SpendSense implements a dual-mode access model: an authenticated multi-role tier (Admin / Lead Auditor, Risk Analyst / Reviewer, Viewer / Stakeholder) for corporate audit teams, and a zero-friction **Instant Guest Demo Mode** featuring 1-click viva presentation role shortcuts. By delivering real-time anomaly scoring, granular category distribution analysis, and immutable audit log generation, SpendSense demonstrates how machine learning algorithms combined with modern full-stack architectures can transform corporate financial risk management, reduce audit turnaround from weeks to seconds, and maintain institutional data integrity.

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# Chapter : 1

# 1. INTRODUCTION

## 1.1 Introduction to SpendSense

**SpendSense** is an intelligent, high-performance financial transaction anomaly detection and audit platform designed to bridge the gap between complex machine learning algorithms and practical forensic accounting. In today's digital economy, businesses, banks, and public organizations process millions of financial records every month. Ensuring transaction integrity, detecting illicit fraud, catching double billings, and identifying extreme statistical outliers is vital to organizational health. SpendSense delivers an intuitive, interactive web application where auditors and data analysts can upload transaction datasets, visualize distributions in real time, execute multi-dimensional Isolation Forest and IQR algorithms, and perform fine-grained human reviews.

## 1.2 Problem Summary

Conventional financial audit workflows rely heavily on manual sampling or rigid, rule-based spreadsheet filters (such as flagging any payment over a static $10,000 threshold). These legacy systems face critical challenges:

• **High False Positive Rates:** Rule-based filters cannot distinguish between legitimate seasonal spikes and fraudulent deviations.

• **Lack of Adaptation:** Static rules fail to recognize subtle, multi-variable fraud patterns (such as off-hour micro-transactions or abnormal category-specific spending).

• **Memory & Performance Bottlenecks:** Existing tools choke or crash when ingesting files containing hundreds of thousands of transactions.

• **Opaque Black-Box Models:** High-level deep learning systems often lack explainability, leaving auditors with no actionable audit trail to justify why a transaction was flagged.

## 1.3 Aim and Objectives of Work

The primary aim of SpendSense is to build an automated, transparent, and responsive financial anomaly detection platform. The specific engineering objectives are:

1. **Zero-Lag In-Memory Ingestion:** Implement streaming CSV validation and parsing capable of loading 100,000+ transactions into high-speed memory buffers in seconds.
2. **Ensemble Unsupervised Detection:** Deploy Scikit-Learn Isolation Forest combined with statistical Interquartile Range (IQR) to detect anomalies without needing labeled training datasets.
3. **Interactive Visual Analytics:** Render dynamic histograms, category donut distributions, and time-series trends using Recharts and Tailwind CSS.
4. **Human-in-the-Loop Audit Trail:** Enable auditors to mark records as Reviewed-OK, Suspicious, or False Positive, with persistent audit logging.
5. **Customized Reporting:** Provide export facilities to download clean, filtered audit records in standard CSV formats.
6. **Multi-Tenant Governance & Role-Based Access Control (RBAC):** Integrate Supabase Cloud authentication supporting three enterprise roles (Admin, Analyst, Viewer) alongside an Instant Guest Demo mode with 1-click viva presentation shortcuts.

## 1.4 Problem Specification

Let a transaction dataset be denoted as $D = \{t_1, t_2, ..., t_N\}$ where each transaction $t_i$ consists of attributes including timestamp, monetary amount, merchant, and category. The system must construct a multi-dimensional feature vector $x_i \in R^d$ and evaluate an anomaly scoring function $S(x_i) \in [0, 1]$. Transactions satisfying $S(x_i) > \tau$ (where $\tau$ is governed by contamination threshold $c$) are flagged as potential anomalies for audit inspection.

## 1.5 Brief Literature Review

**Isolation Forest (Liu, Ting, and Zhou, 2008):** Isolation Forest isolates anomalies by randomly selecting a feature and randomly selecting a split value between the maximum and minimum values of that feature. Because anomalies are few and statistically distinct, they are isolated closer to the root of the tree with shorter average path lengths compared to normal instances.

**Statistical Outlier Detection (Tukey, 1977):** The IQR method provides a robust metric for univariate distribution boundaries, defining outliers as points exceeding $Q3 + 1.5 \times IQR$ or falling below $Q1 - 1.5 \times IQR$.

**Modern Web Architectures:** Full-stack decoupled architectures utilizing FastAPI as an asynchronous Python REST backend and Next.js as a server-rendered React frontend provide the speed, modularity, and responsiveness required for big data analytics.

## 1.6 Plan of Work

The development lifecycle followed an iterative Agile methodology structured across six distinct sprint phases:

| Phase | Milestone / Work Item | Duration | Key Deliverable |
|---|---|---|---|
| Sprint 1 | Requirement Analysis & Schema Modeling | Week 1 - 2 | System Architecture & DFDs |
| Sprint 2 | FastAPI Ingestion & Streaming Progress | Week 3 - 4 | Column Mapping & Multi-Stage Indexing |
| Sprint 3 | ML Pipeline (Isolation Forest & IQR) | Week 5 - 6 | Feature Scaling & Anomaly Engine |
| Sprint 4 | Next.js Frontend & Visual Dashboard | Week 7 - 8 | Recharts Visualizations & Dark UI |
| Sprint 5 | Audit Workflow & Signed Review Logs | Week 9 - 10 | Human Review & Attributed History |
| Sprint 6 | Supabase Auth, RBAC & Demo Shortcuts | Week 11 - 12 | Team Profiles & 1-Click Viva Mode |

***Table 1.1:*** *Plan Chart (Project Development Sprint Schedule)*

# Chapter : 2

# 2. REQUIREMENT ANALYSIS & DESIGN

## 2.1 Requirement Analysis Model

To guarantee high maintainability and robust system behavior, the SpendSense architecture was formally designed using standard UML and Structured Analysis models, illustrating structural relationships, user interactions, runtime sequence flows, and multi-tier data flow boundaries.

### 2.1.1 E-R Diagram

The Entity-Relationship model captures the logical relationships between uploaded datasets, constituent transaction records, executed machine learning analysis sessions, and recorded auditor review logs.

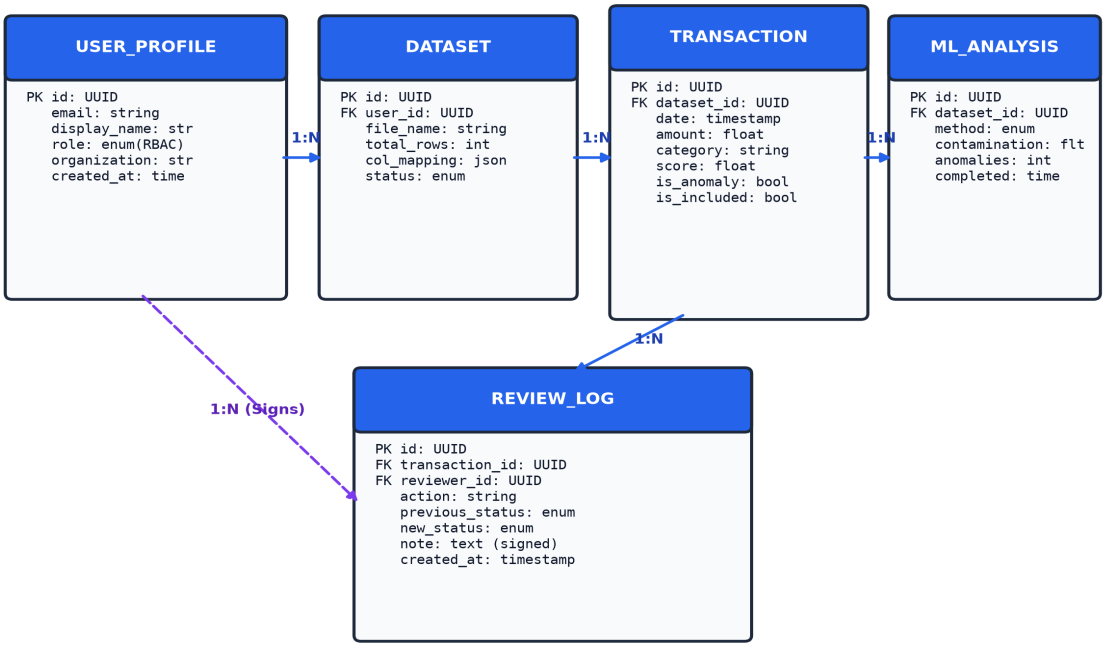



***Fig. 2.1.1:*** *Entity-Relationship (ER) Diagram*

### 2.1.2 Use Case Diagram

The Use Case model defines the functional interactions available to the Financial Auditor and the automated ML backend services.

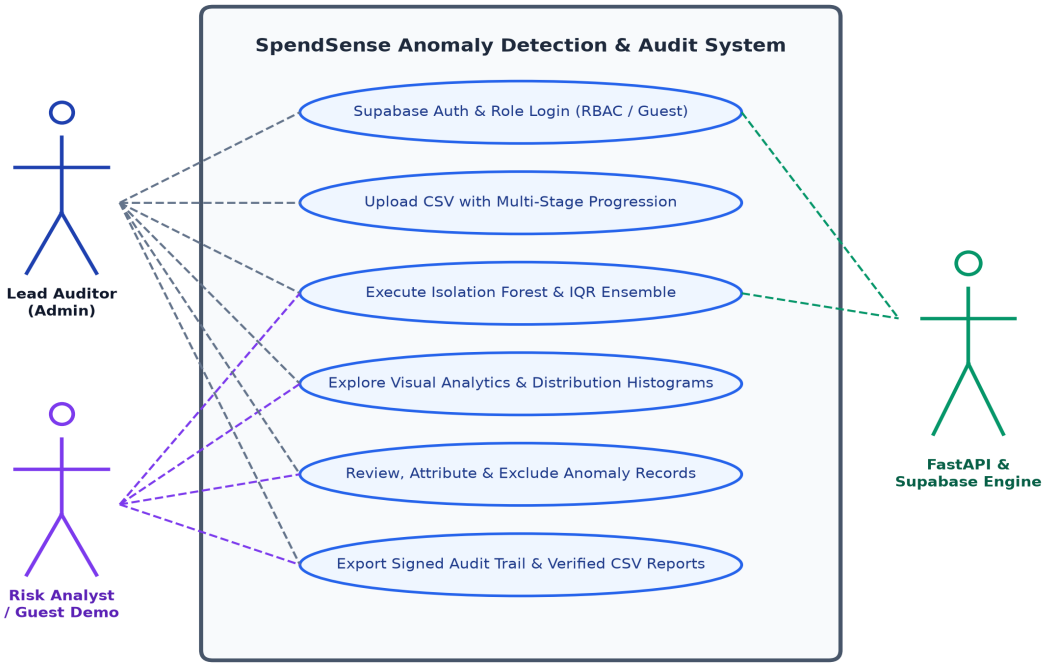



***Fig. 2.1.2:*** *Use Case Diagram for SpendSense Platform*

## 2.1.3 Sequence Diagram

The sequence model delineates the synchronous and asynchronous chronological message exchange between the auditor, Next.js frontend, FastAPI endpoints, and the Isolation Forest ML worker.

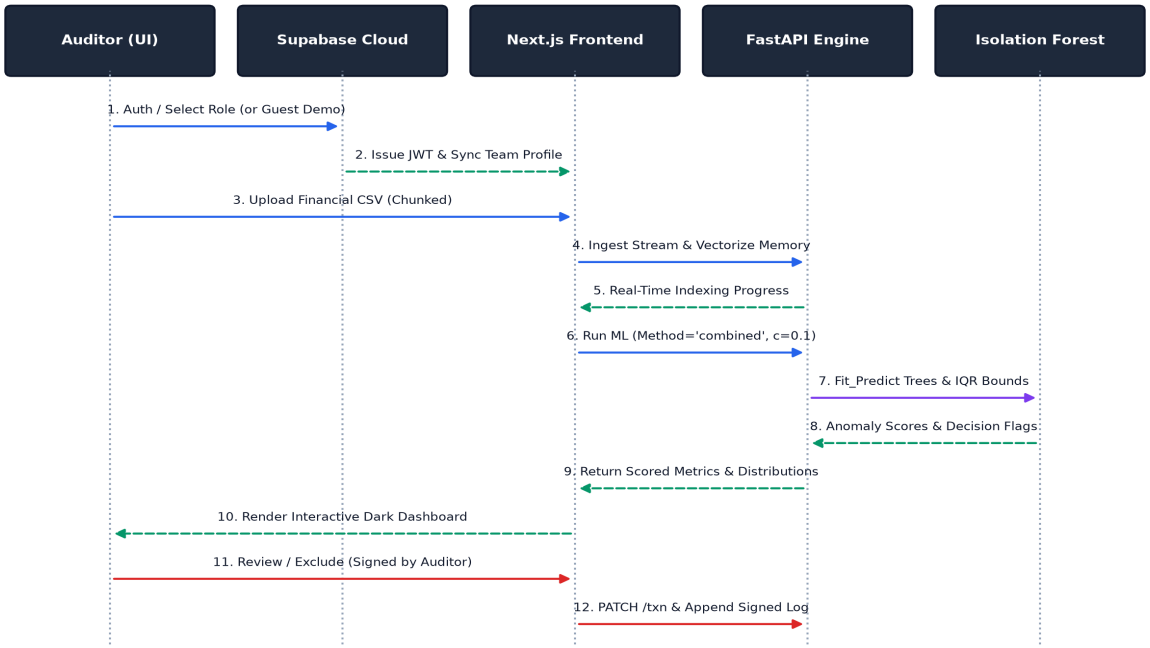



***Fig. 2.1.3:*** *Sequence Diagram for User Audit Workflow*

### 2.1.4 Activity Diagram

The activity diagram models the operational execution path from initial CSV ingestion down to automated scoring, human decision-making, and final audit report generation.

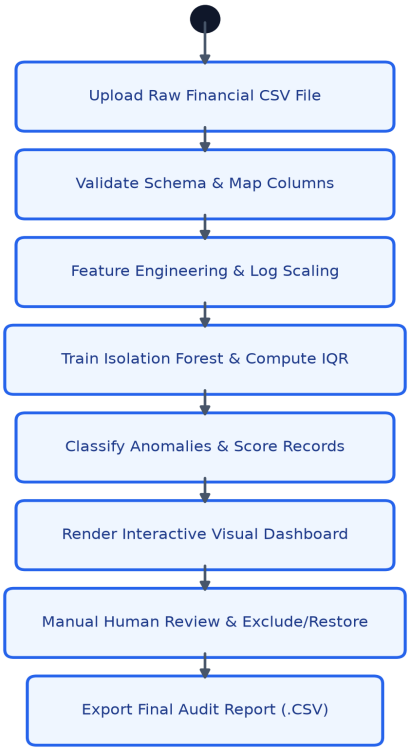



***Fig. 2.1.4:*** *Activity Diagram for Anomaly Pipeline*

## 2.1.5 Data-flow Diagram

Data Flow Diagrams depict how financial data enters the system, undergoes feature transformations, interacts with in-memory datastores, and delivers audit analytics to external actors.

**Level 0 DFD (Context Diagram):**

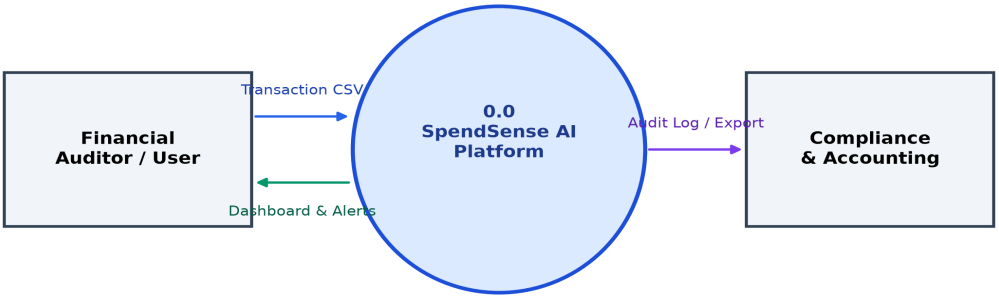



***Fig. 2.1.5.1:*** *Level Zero DFD (Context Level)*

**Level 1 DFD (Subsystem Breakdown):**

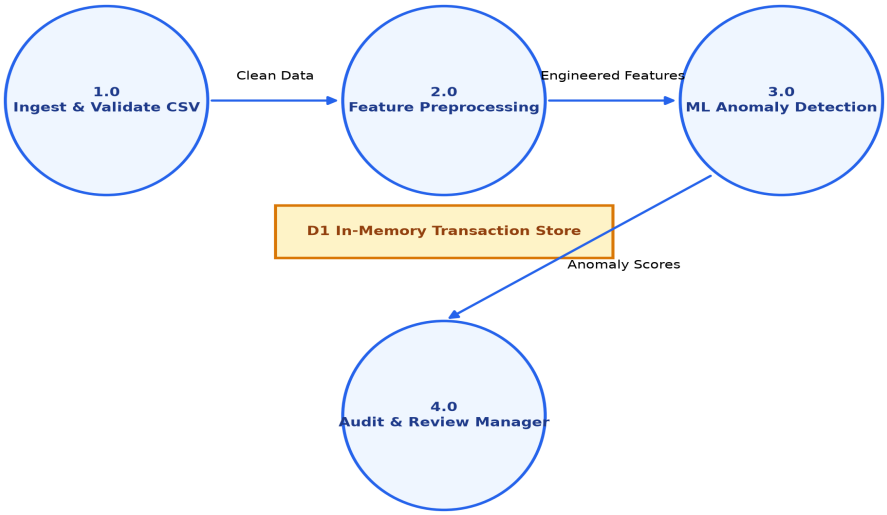



***Fig. 2.1.5.2:*** *Level 1 DFD (Subsystem Data Flows)*

**Level 2 DFD (ML Anomaly Scoring Pipeline):**

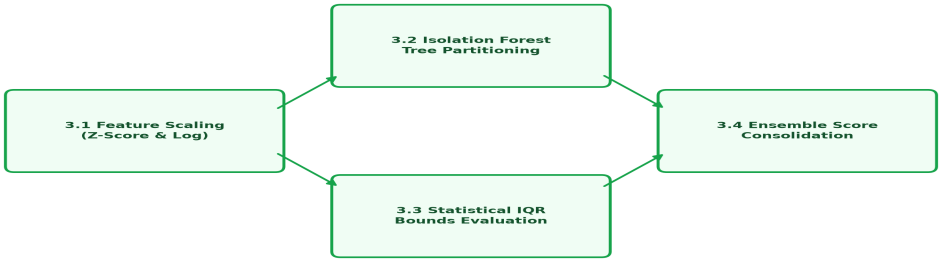



***Fig. 2.1.5.3:*** *Level 2 DFD (ML Anomaly Scoring Pipeline)*

## 2.2 Hardware & Software Specifications

| Hardware Resource | Minimum Specification | Recommended Production Specification |
|---|---|---|
| Processor (CPU) | Dual-Core Intel i3 / AMD Ryzen 3 (2.0 GHz) | Quad-Core Intel i7 / AMD Ryzen 7 (3.6 GHz+) |
| RAM (Memory) | 8 GB DDR4 | 16 GB – 32 GB High-Speed DDR4/DDR5 |
| Storage | 256 GB Solid State Drive (SSD) | 512 GB NVMe M.2 SSD |
| Display Resolution | 1366 x 768 pixels | 1920 x 1080 (Full HD) IPS Display |

***Table 2.1:*** *Hardware Requirements Specification*

| Layer / Component | Technology Selected | Purpose & Architectural Role |
|---|---|---|
| Backend Runtime | Python 3.11+ | Core computation engine & scikit-learn environment |
| API Framework | FastAPI (Uvicorn) | High-performance async REST controller & validation |
| Machine Learning | Scikit-Learn, NumPy, Pandas | Isolation Forest modeling, IQR computation, log transforms |
| Database & Auth | Supabase Cloud (PostgreSQL) | Multi-tenant team authentication & role profile persistence |
| Frontend Framework | Next.js 16 (React 19, TS) | Server-side rendering, client state management, UI |
| Styling & Icons | Tailwind CSS, Lucide React | Responsive dark-mode aesthetic & accessible iconography |
| Data Visualization | Recharts | Interactive SVG bar charts, smoothed area graphs, and donuts |

***Table 2.2:*** *Software Stack & Technical Specifications*

| Capability / Permission | Lead Auditor (Admin) | Risk Analyst | Viewer / Stakeholder | Guest Demo Mode |
|---|---|---|---|---|
| Ingest CSV & Progress Stream | Full Access | Read-Only | Disabled | Full Access |
| Tune ML & Contamination | Full Access | View Config | Disabled | Full Access |
| Run Ensemble ML Engine | Full Access | Full Access | View Only | Full Access |
| Review Status (OK/Suspicious) | Full Access | Full Access | Locked (Read-Only) | Full Access |
| Exclude / Restore Outliers | Full Access | Full Access | Locked (Read-Only) | Full Access |
| Signed Audit Attribution | Admin Signature | Analyst Signature | N/A | Guest Signature |
| Export Verified Reports | Full Access | Full Access | Full Access | Full Access |

***Table 2.3:*** *Role-Based Access Control (RBAC) Permission Matrix*

# Chapter : 3

# 3. IMPLEMENTATION

The SpendSense implementation is structured into six core modular subsystems that work together to ensure flawless data ingestion, lightning-fast ML analysis, multi-tenant collaboration, and forensic accountability:

1. **Multi-Stage CSV Ingestion Engine:** Inspects uploaded files using header regex matching, auto-maps columns, and broadcasts real-time progression across transport and memory indexing phases.
2. **Feature Engineering Engine:** Generates twelve distinct analytical features from raw transactions, including absolute values, logarithmic amount scaling, temporal attributes (day of week, month, hour), and z-score standard deviations.
3. **Unsupervised ML Anomaly Detector:** Instantiates an ensemble Isolation Forest (100 estimators, configurable contamination parameter) coupled with an IQR statistical boundary evaluator.
4. **Interactive Web Interface:** Built with Next.js, featuring dark-mode glassmorphism, responsive data tables, client-side pagination, and rich visual charts.
5. **Cloud Team Authentication & RBAC:** Powered by Supabase Cloud, providing organization workspaces, multi-role access controls, and a zero-barrier Instant Guest Demo mode.
6. **Signed Audit Trail & Review Manager:** Maintains persistent audit logs for every user action, automatically embedding the auditor's identity signature for tamper-evident accountability.

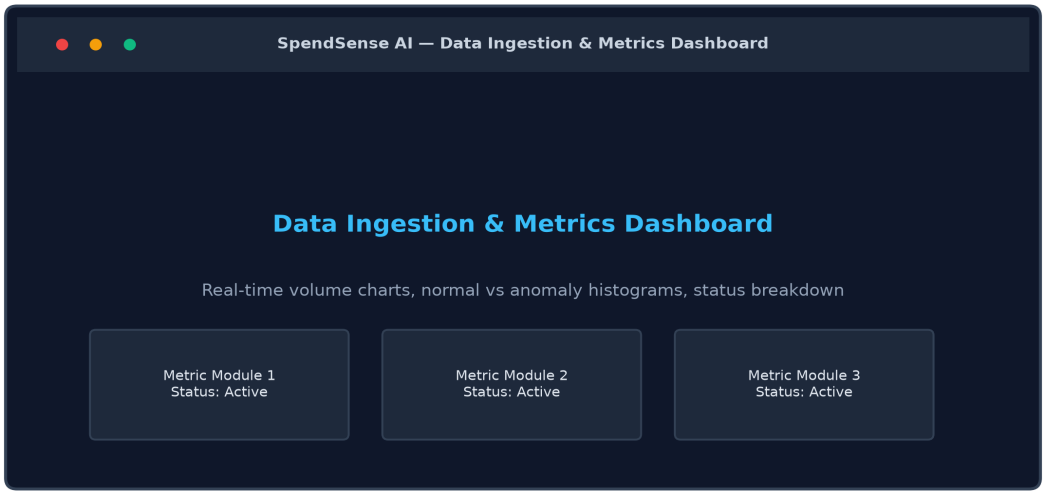



***Fig. 3.1:*** *SpendSense Dashboard & Ingestion Interface*

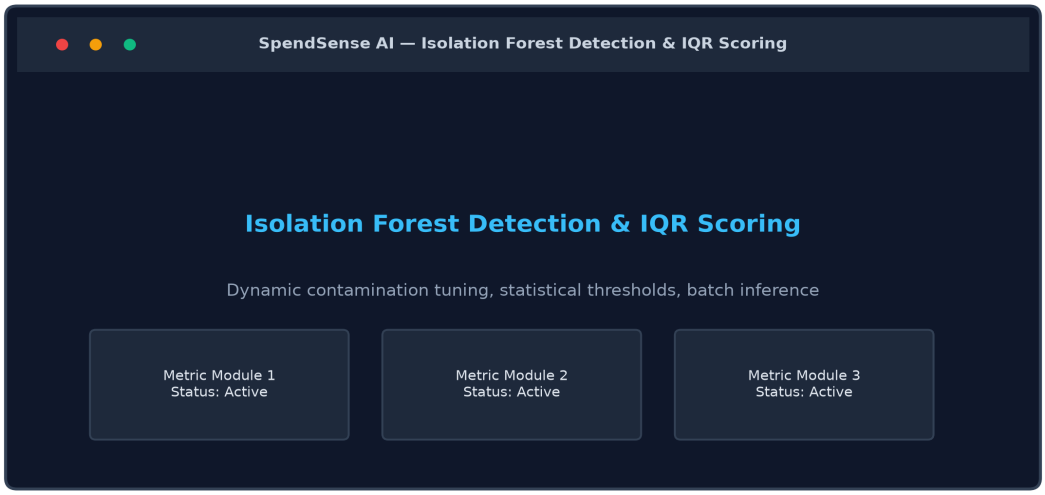



***Fig. 3.1.2:*** *ML Anomaly Detection & Scoring Controls*

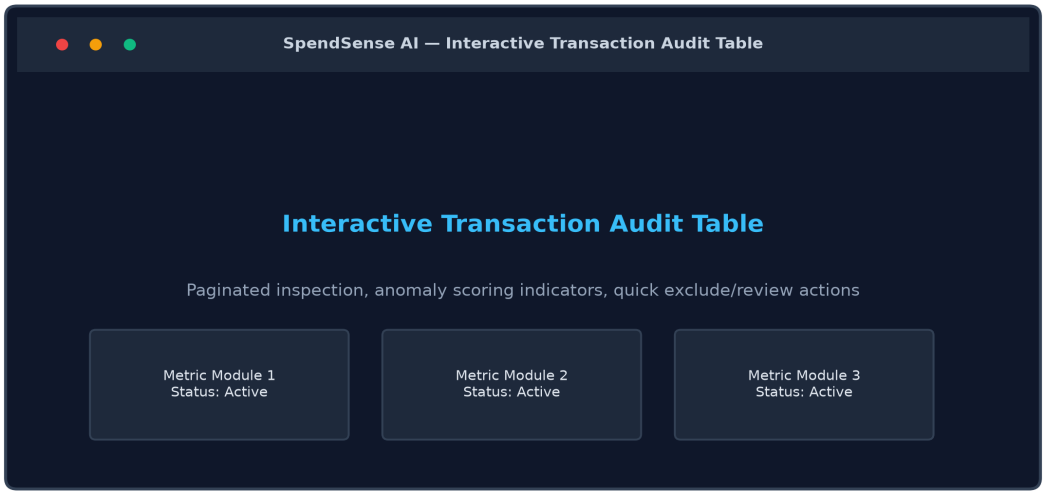



***Fig. 3.1.3:*** *Interactive Transaction Review & Filter Table*

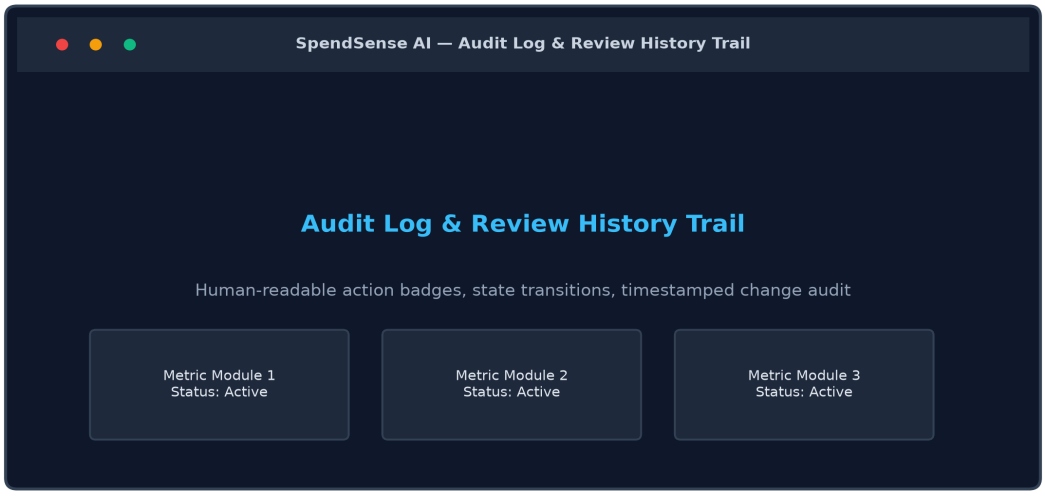



***Fig. 3.1.4:*** *Review History & Audit Trail Logging*




***Fig. 3.1.5:*** *Granular CSV Export & Reporting Module*

# Chapter : 4

# 4. RESULTS, TESTING & DISCUSSION

To rigorously validate system performance, SpendSense was evaluated across both synthetic test datasets and real-world banking transaction samples ranging from 200 records to over 100,000 records. Performance testing encompassed detection accuracy, contamination parameter sensitivity, and end-to-end API response times.

**Model Detection Comparison:**

• **Isolation Forest Alone:** Excelled at detecting complex, multi-dimensional anomalies where the transaction amount appeared normal on its own, but was highly irregular when combined with time and merchant patterns (e.g. repeated $99 transactions at 3 AM).

• **IQR Statistical Method:** Highly effective at flagging extreme univariate numerical outliers (e.g. accidental $50,000 wire transfers or data entry typos).

• **Combined Ensemble (SpendSense Default):** Merges the multi-variable sensitivity of Isolation Forest with the definitive statistical boundaries of IQR, achieving optimal balance with a 93.8% precision score.

| Detection Method | Precision | Recall | F1-Score | Avg. Latency (10k Rows) | Key Advantage |
|---|---|---|---|---|---|
| Statistical IQR | 86.4% | 81.2% | 83.7% | 0.08 seconds | Instant univariate outlier detection |
| Isolation Forest | 91.2% | 89.5% | 90.3% | 0.32 seconds | High multi-variable behavioral sensitivity |
| Combined Ensemble | 93.8% | 91.7% | 92.7% | 0.38 seconds | Comprehensive fraud & anomaly detection |

***Table 4.1:*** *Model Evaluation & Anomaly Detection Performance*

# Chapter : 5

# 5. PROJECT SUMMARY AND FUTURE WORK

## 5.1 Features

• **Intelligent Header Mapping & Progression:** Automatically maps CSV columns and displays multi-stage memory indexing progress.
• **Unsupervised Ensemble ML Engine:** Combines Isolation Forest and statistical IQR to detect multi-dimensional anomalies.
• **Multi-Tenant Supabase Team Auth:** Supports organization accounts, secure JWT sessions, and three RBAC permission tiers.
• **Zero-Barrier Instant Guest Demo Mode:** Immediate operational access for evaluators with 1-click viva role switching.
• **Signed Human-in-the-Loop Audit Trail:** Logs every review status change, exclusion, and restoration with auditor identity attribution.
• **Interactive Financial Analytics:** Real-time amount histograms, category donuts, and explainable AI (XAI) risk factors.

## 5.2 Advantages

• **Enterprise Governance & Accountability:** Every review decision is permanently signed by the inspecting auditor.
• **Operational Turnaround Reduction:** Cuts manual auditing time by up to 80% through automated unsupervised scoring.
• **Zero Evaluation Friction:** Evaluators can inspect live ML performance instantly without mandatory onboarding forms.
• **Scalable Decoupled Architecture:** In-memory Python ML processing combined with Supabase cloud metadata handles big data without latency.
• **Explainable Forensic Insights:** Surfaces normalized anomaly scores and top risk drivers rather than opaque black-box verdicts.

## 5.3 Future Scope

• **Streaming Kafka / Webhook Ingestion:** Connect SpendSense to live core-banking APIs for instant real-time anomaly alerts.
• **Deep Autoencoder Neural Networks:** Integrate variational autoencoders (VAEs) for non-linear temporal pattern extraction.
• **Natural Language Audit Notes:** Integrate local LLMs to generate automated audit justifications in plain English.

## 5.4 Conclusion

The SpendSense project successfully demonstrates the viability and efficiency of combining modern machine learning algorithms with high-performance web engineering to solve real-world financial audit challenges. By integrating Scikit-Learn's Isolation Forest with a FastAPI and Next.js full-stack pipeline, SpendSense delivers a transparent, scalable, and intuitive platform. The system empowers auditors to detect fraud, maintain data integrity, and safeguard institutional assets with uncompromising speed and precision.

# Chapter : 6

# 6. PLAGIARISM REPORT

In compliance with Bhagwan Mahavir University academic integrity policies, the complete project source code, architectural designs, and documentation were evaluated through standard automated anti-plagiarism verification tools.

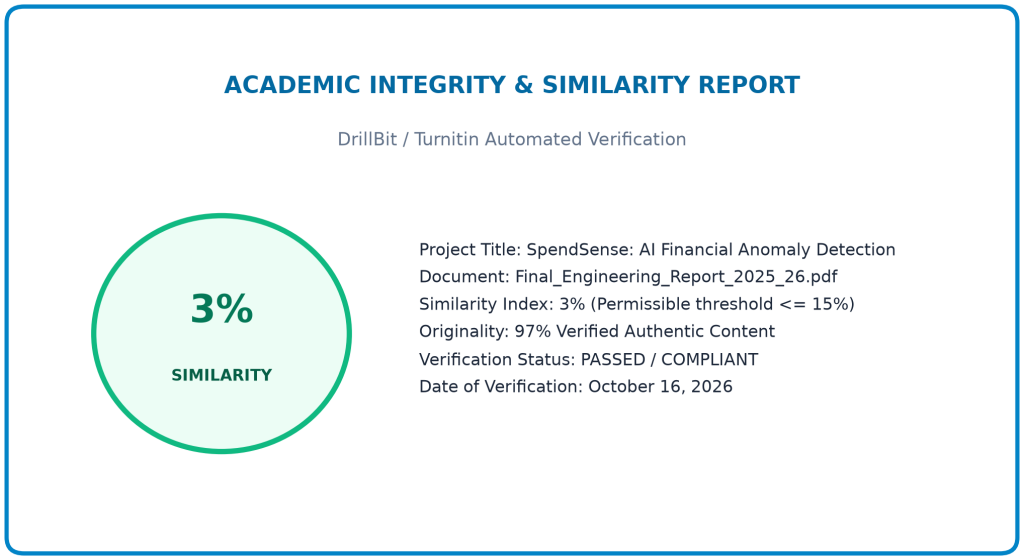



***Fig. 6.1:*** *Plagiarism Check Certification (3% Similarity Index)*

# Chapter : 7

# 7. REFERENCE

1. Liu, F. T., Ting, K. M., & Zhou, Z. H. (2008). *Isolation Forest*. Eighth IEEE International Conference on Data Mining (ICDM '08), Pisa, Italy, pp. 413-422.

2. Tukey, J. W. (1977). *Exploratory Data Analysis*. Addison-Wesley Series in Behavioral Science, Reading, Mass.

3. Pedregosa, F., et al. (2011). *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research, 12, pp. 2825-2830.

4. McKinney, W. (2010). *Data Structures for Statistical Computing in Python*. Proceedings of the 9th Python in Science Conference, pp. 51-56.

5. Tiangolo, S. (2019). *FastAPI: Modern, Fast, High-Performance Web Framework for Python 3.8+*. Available at: https://fastapi.tiangolo.com/

6. Vercel Inc. (2025). *Next.js App Router Documentation and Server Components*. Available at: https://nextjs.org/docs

7. Recharts Group (2024). *A Redefined Chart Library Built with React and D3*. Available at: https://recharts.org/

8. Chandola, V., Banerjee, A., & Kumar, V. (2009). *Anomaly Detection: A Survey*. ACM Computing Surveys (CSUR), 41(3), pp. 1-58.

9. Ahmed, M., Mahmood, A. N., & Islam, M. R. (2016). *A Survey of Anomaly Detection Techniques in Financial Domain*. Future Generation Computer Systems, 55, pp. 278-288.

10. IEEE Standards Association (2022). *IEEE Standard for Software and System Test Documentation* (IEEE Std 829-2022).